

Guidelines for importing and using the SNA image file with VirtualBox

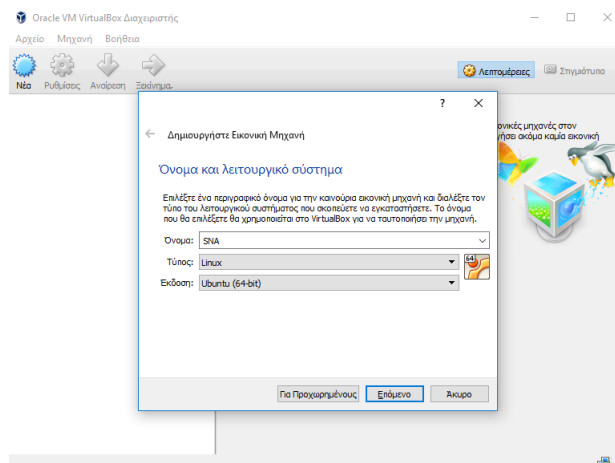
INF322: Social Network Analysis
Athens University of Economic and Business



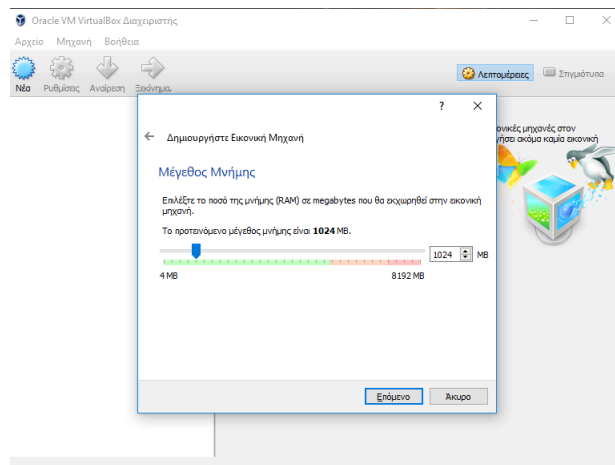
- ☞ This document assumes that virtualization is supported by your system and is already enabled. Depending on your operating system you can verify if that is the case with different free tools; Speccy^a is one such option for Microsoft Windows users.

^a<https://www.piriform.com/speccy>

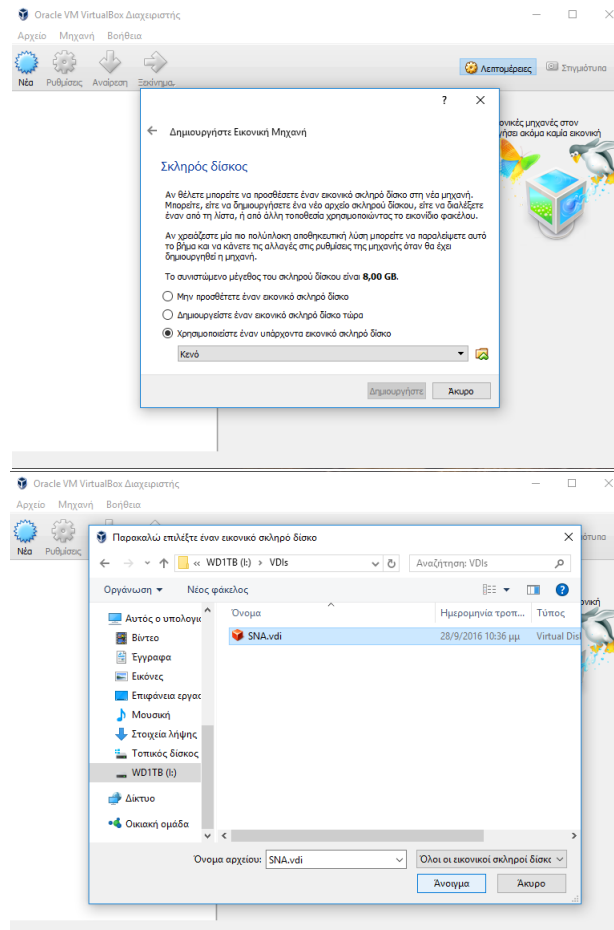
- Download and install VirtualBox from <https://www.virtualbox.org/wiki/Downloads>.
- Start VirtualBox and create a **new** virtual machine (Linux, Ubuntu 64-bit).



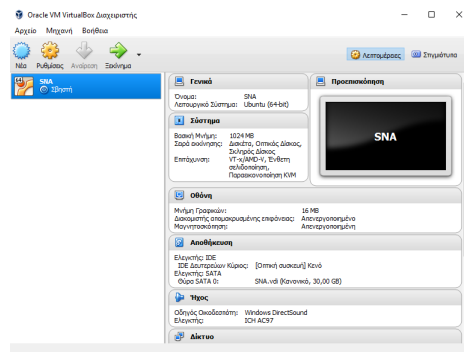
- Set the memory size. A size of 512MB would suffice for most tasks. However, if you plan to use the Eclipse Neon IDE that is preinstalled you should use at least 2048MB for a comfortable user experience.



- When you're asked for a hard disk image, select 'Use existing hard disk' and click on the small icon on the right to select the .vdi image provided for this class. The URL for the image file is: <https://goo.gl/CGm1YR>.



- Back in the main window, you're now able to start your new virtual machine.

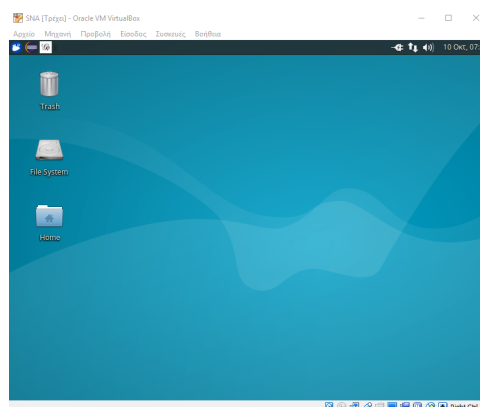


- Your virtual machine runs Xubuntu 16.04. Two users are available:
 - username: `sna` (default user)
password: `mysn@vm`
 - username: `hduser` (user for executing hadoop jobs)
password: `h@doop`

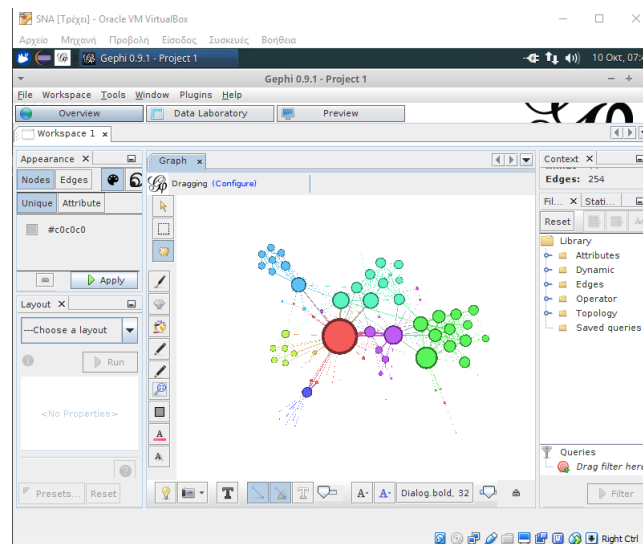
The VM has preinstalled:

- Oracle Java 8
- gephi 0.9.1
- Snap 3.0
- Apache Giraph 1.1.0 with Hadoop 0.20.203.0
- Eclipse Neon IDE

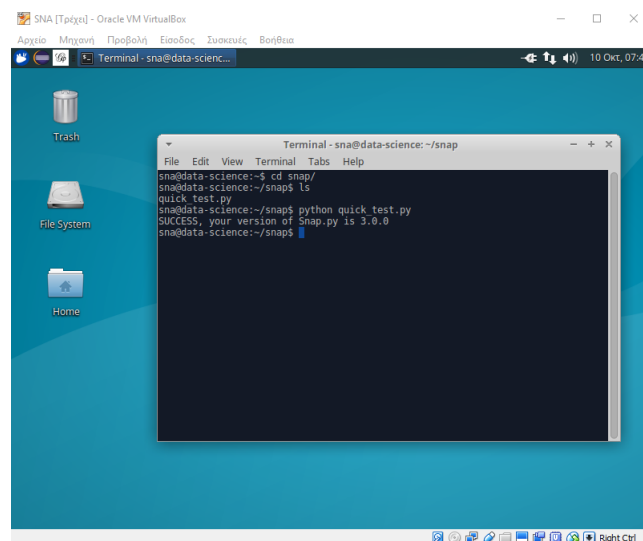
The disk size may reach 30GB out of which 1GB is dedicated to a swap partition.



- You can start **Gephi** from the launcher on the taskbar at the top of your screen.

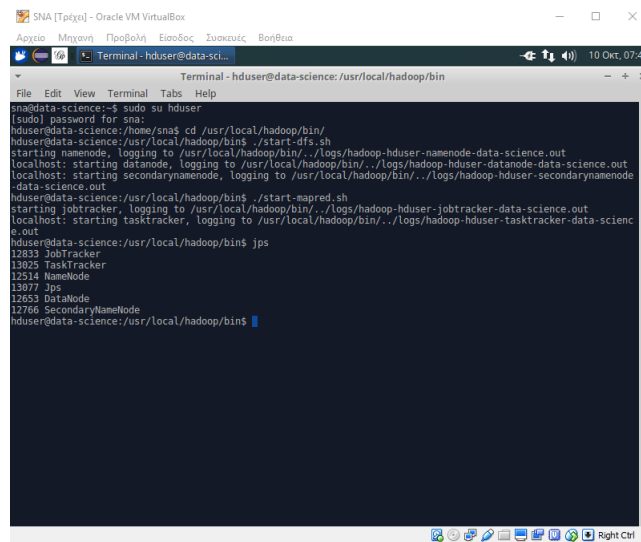


- A python script verifying the presence of the **Snap** library is located at the `'/home/sna/snap'` directory.



- Executing Apache Giraph jobs requires starting HDFS and MAP-REDUCE. You should **always** switch user to **hduser** to start both. Then you may submit a Giraph job, again as **hduser**. Running **jps** allows you to verify that initialization was error free. In particular, all the following should be running:

- NameNode
- SecondaryNameNode
- DataNode
- JobTracker
- TaskTracker



```
SNA [Τρέχει] - Oracle VM VirtualBox
Αρχείο  Μενю  Προβολή  Είσοδος  Συνδεθείτε  Βοήθεια
Terminal - hduser@data-sci...  10 Oct, 07:46
File Edit View Terminal Tabs Help
snagdata-science:~$ sudo su hduser
(sudo) password for sna:
hduser@data-science:/home/sna$ cd /usr/local/hadoop/bin/
hduser@data-science:/usr/local/hadoop/bin$ ./start-dfs.sh
starting namenode, logging to /usr/local/hadoop/bin/../logs/hadoop-hduser-namenode-data-science.out
localhost: starting datanode, logging to /usr/local/hadoop/bin/../logs/hadoop-hduser-datanode-data-science.out
localhost: starting secondarynamenode, logging to /usr/local/hadoop/bin/../logs/hadoop-hduser-secondarynamenode-data-science.out
hduser@data-science:/usr/local/hadoop/bin$ ./start-mapred.sh
starting jobtracker, logging to /usr/local/hadoop/bin/../logs/hadoop-hduser-jobtracker-data-science.out
localhost: starting tasktracker, logging to /usr/local/hadoop/bin/../logs/hadoop-hduser-tasktracker-data-science.out
hduser@data-science:/usr/local/hadoop/bin$ jps
12833 JobTracker
13025 TaskTracker
12514 NameNode
13077 Jps
12653 DataNode
12786 SecondaryNameNode
hduser@data-science:/usr/local/hadoop/bin$
```

- Firefox has two bookmarks in its toolbar to the NameNode and the Job-Tracker. With these interfaces you can browse the HDFS or see running jobs.

The screenshot shows a Firefox browser window with the address bar set to `localhost:50070/dfshealth.jsp`. The page title is "NameNode 'localhost:54310'". The main content area displays the following information:

- Started:** Mon Oct 10 07:45:55 EEST 2016
- Version:** 0.20.203.0, r1099333
- Compiled:** Wed May 4 07:57:50 PDT 2011 by oom
- Upgrades:** There are no upgrades in progress.

Below this information are links for [Browse the filesystem](#) and [NameNode Logs](#). A "Cluster Summary" section follows, showing:

- 42 files and directories, 6 blocks = 48 total. Heap Size is 27.05 MB / 966.69 MB (2%)
- Configured Capacity : 28.46 GB
- DFS Used : 100 KB
- Non DFS Used : 7.91 GB
- DFS Remaining : 20.55 GB
- DFS Used% : 0 %
- DFS Remaining% : 72.21 %

The screenshot shows a Firefox browser window with the address bar set to `localhost:50030/jobtracker.jsp`. The page title is "localhost Hadoop Map/Reduce Administration". The main content area displays the following information:

- State:** RUNNING
- Started:** Mon Oct 10 07:46:44 EEST 2016
- Version:** 0.20.203.0, r1099333
- Compiled:** Wed May 4 07:57:50 PDT 2011 by oom
- Identifier:** 201610100746

Below this information is a "Cluster Summary (Heap Size is 15.5 MB/966.69 MB)" section, which includes a table with the following data:

Running Map Tasks	Running Reduce Tasks	Total Submissions	Nodes	Occupied Map Slots	Occupied Reduce Slots	Reserved Map Slots	Reserved Reduce Slots	Max Task Capacity
0	0	0	1	0	0	0	0	4

- **Always** remember to terminate MAP-REDUCE and HDFS (in this order) before shutting down your VM.

```

SNA [Tpxen] - Oracle VM VirtualBox
Αρχείο Μενού Προβολή Εισόδος Έξοδος Βοήθεια
Terminal - hduser@data-sci...
Terminal - hduser@data-sci: /usr/local/hadoop/bin
File Edit View Terminal Tabs Help
hduser@data-sci: /usr/local/hadoop/bin$ ./stop-mapred.sh
stopping jobtracker
localhost: stopping tasktracker
hduser@data-sci: /usr/local/hadoop/bin$ ./stop-dfs.sh
stopping namenode
localhost: stopping datanode
localhost: stopping secondarynamenode
hduser@data-sci: /usr/local/hadoop/bin$ jps
13940 jps
hduser@data-sci: /usr/local/hadoop/bin$

```

- The **Eclipse Neon IDE** is installed and may be accessed by a launcher in the top taskbar. **Apache Giraph** is imported as a **Maven** project. You can use this IDE or any editor of your choice to write Java code. A more lightweight option is **gedit** that can be installed by issuing the command: `sudo apt-get install gedit`.

